AIR HEAT EXCHANGER - DESIGN OPTIMIZATION SUMMARY

Generated: 12-Apr-2026

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FIXED OPERATING CONDITIONS:

  Air mass flow rate:       0.0570722222 kg/s
  Hot fluid mass flow:      0.25 kg/s (Helisol 5A)
  Air inlet temperature:    25°C
  Air outlet temperature:   280°C
  Salt inlet temperature:   330°C
  Heat duty:                14.76 kW
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DESIGN COMPARISON

Parameter	Original	V1 (Min DP)	V2 (Compact)	Units
Shell Length	2000	3000	1750	mm
Shell Diameter	238	189	211	mm
L/D Ratio	8.4	15.9	8.3	-
Shell Volume	~89	~84	61.4	L
Tube-side DP	7.00	0.24	0.51	kPa
Shell-side DP	0.068	0.049	0.054	Pa
Number of Tubes	24	14	12	-
Tube Passes	4	2	2	-
Tube ID	10	18	15	mm
Tube OD	14	20	18	mm
Fin Height	8	5	10	mm
Fin Pitch	3.0	2.0	2.0	mm
Overall HTC	6.38	4.77	4.22	W/m²/K

DESIGN DESCRIPTIONS:

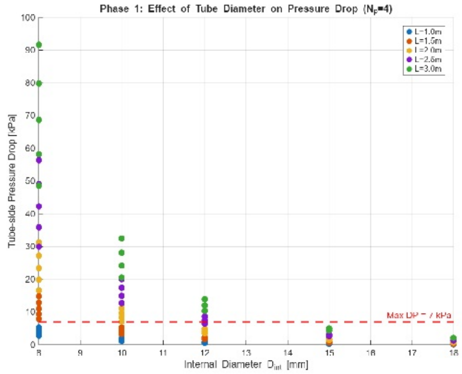
Original Baseline: The starting design with 4 tube passes and 10mm ID tubes. Pressure drop at the 7 kPa limit. Standard engineering approach.

V1 (Minimum Pressure Drop): Optimized primarily to minimize tube-side pressure drop. Uses larger tubes (18mm ID) and fewer passes (2). Longer but narrower shell. Achieved 97% reduction in pressure drop.

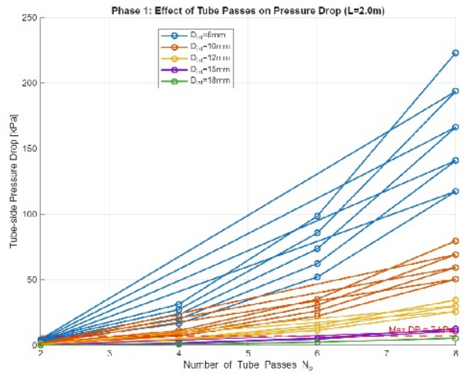
V2 (Compact Design): Optimized to minimize shell volume while meeting all pressure constraints. Balances length and diameter for smallest overall footprint. Achieved 31% volume reduction vs original.

V1 OPTIMIZATION: MINIMIZING PRESSURE DROP

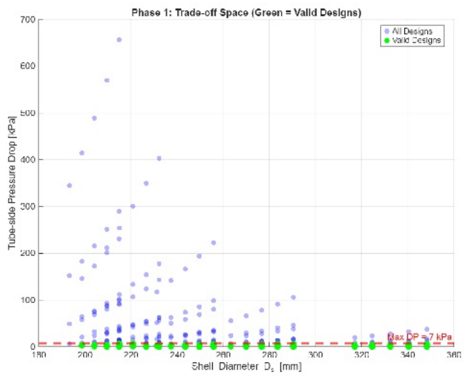
Effect of Tube Diameter



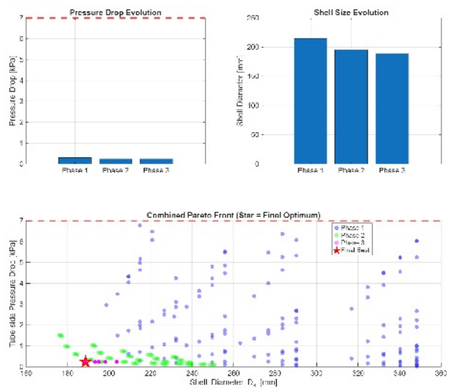
Effect of Tube Passes



Phase 1 Trade-off Space



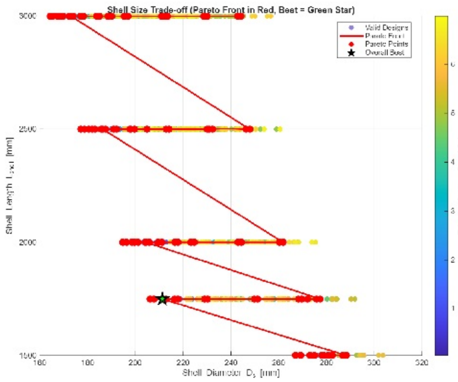
Optimization Progress



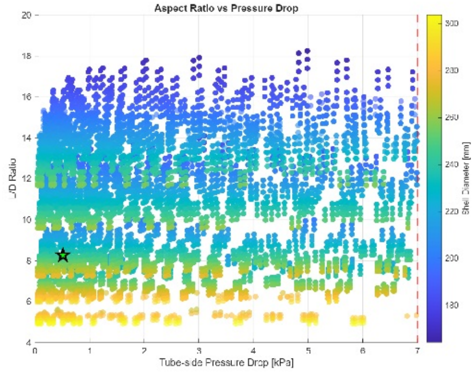
KEY FINDINGS - V1:
- Tube internal diameter is the dominant factor - increasing from 10mm to 18mm reduced DP by 97%
- Number of passes has near-linear effect on pressure drop - 2 passes vs 4 passes cuts DP by ~75%
- Trade-off: Longer shell (3m) required to compensate for fewer, larger tubes

V2 OPTIMIZATION: COMPACT DESIGN (MINIMUM VOLUME)

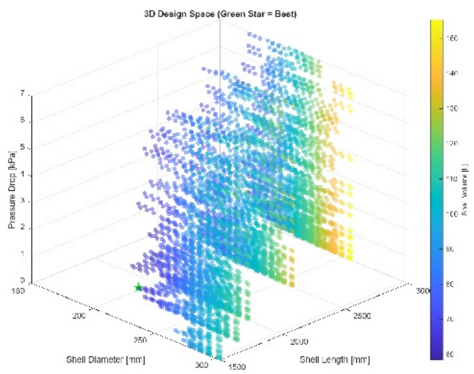
Shell Size Trade-off



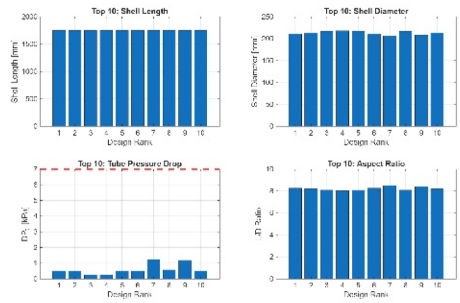
Aspect Ratio vs Pressure Drop



3D Design Space

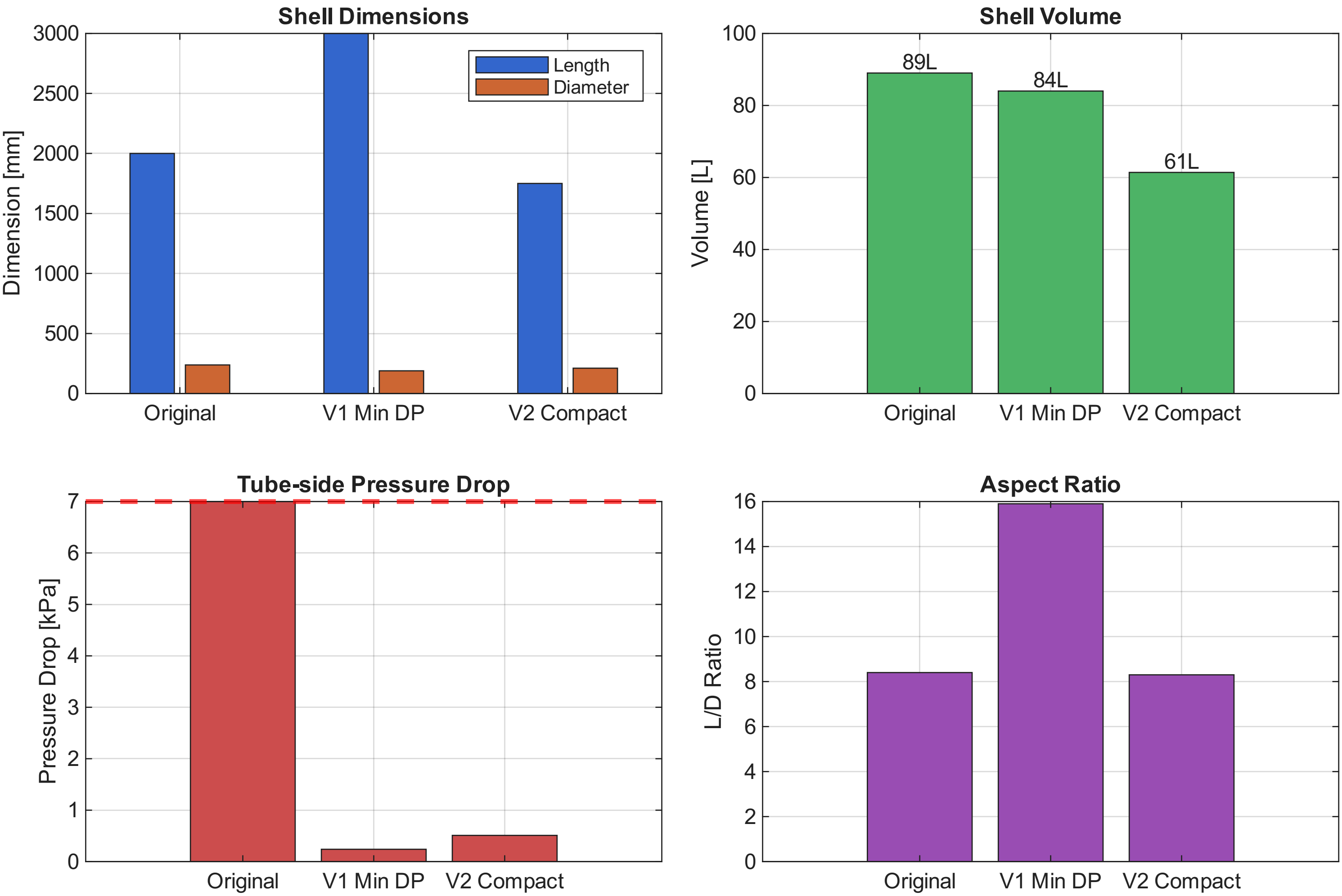


Top 10 Compact Designs



KEY FINDINGS - V2:
- Clear trade-off between shell length and diameter - shorter shells require wider diameter
- Pareto front shows minimum achievable L/D ratio of ~5.5 (at 1500mm length)
- Optimal compact design: 1750mm x 211mm achieves 31% volume reduction while meeting all constraints

DESIGN RECOMMENDATIONS



RECOMMENDATIONS:

Choose V2 COMPACT DESIGN if:
- Space is limited and a smaller footprint is critical
- Installation constraints favor a shorter exchanger
- The slightly higher pressure drop (0.51 kPa) is acceptable

Choose V1 MIN DP DESIGN if:
- Minimizing pumping power is the primary concern
- A longer, narrower exchanger fits the installation space
- Maximum margin on pressure drop constraint is desired

AVOID ORIGINAL DESIGN:
- Operating at the pressure drop limit with no safety margin
- Larger volume than necessary

All optimized designs meet the thermal requirements (280°C air outlet) and pressure constraints (DP_t < 7 kPa, DP_s < 0.068 Pa).